Week-9 Lab Programs

1. Water Jug Problem

AIM

To solve the Water Jug Problem by using two jugs of different capacities and a water source to measure a specific quantity of water. The objective is to reach the desired amount by applying operations such as filling, emptying, and pouring water between the jugs.

Description

The Water Jug Problem is a classic Artificial Intelligence problem that demonstrates state-space search. Two jugs with fixed capacities are used to measure a target quantity of water. The solution is obtained by performing valid operations, including filling, emptying, and transferring water between the jugs, until the goal state is reached.

Code

```
from math import gcd

jug1 = int(input("Enter capacity of Jug 1: "))
jug2 = int(input("Enter capacity of Jug 2: "))
target = int(input("Enter target amount: "))

if (target > max(jug1, jug2) or
        target % gcd(jug1, jug2) != 0):
    print("No solution possible")
else:
    a = 0
    b = 0
    steps = [(a, b)]

    while (a != target or b != 0) and \
          (a != 0 or b != target):

        if a == 0:
            a = jug1
        elif b == jug2:
            b = 0
        else:
            amount = min(a, jug2 - b)
            a -= amount
            b += amount
```

```
        steps.append((a, b))

    print("\\nSteps:")
    for i in range(len(steps)):
        print("Step", i, ":", steps[i])

    print("\\nFinal State:")
    print("Jug 1 =", a)
    print("Jug 2 =", b)
```

Output

```
Enter capacity of Jug 1: 4
Enter capacity of Jug 2: 3
Enter target amount: 2

Steps:
Step 0 : (0, 0)
Step 1 : (4, 0)
Step 2 : (1, 3)
Step 3 : (1, 0)
Step 4 : (0, 1)
Step 5 : (4, 1)
Step 6 : (2, 3)
Step 7 : (2, 0)

Final State:
Jug 1 = 2
Jug 2 = 0
```

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2. Missionaries and Cannibals Problem

AIM

To solve the Missionaries and Cannibals Problem using Artificial Intelligence techniques by safely transporting three missionaries and three cannibals across a river without allowing the cannibals to outnumber the missionaries on either bank when missionaries are present.

Description

The Missionaries and Cannibals Problem is a classic AI problem that demonstrates state-space search. Three missionaries and three cannibals must cross a river using a boat that can carry at most two people. The solution involves finding a sequence of valid moves while ensuring that missionaries are never outnumbered by cannibals on either riverbank.

Code

```
from collections import deque

start = (3, 3, 0)
goal = (0, 0, 1)

def valid(m, c):
    return (m == 0 or m >= c) and \
           (m == 3 or 3 - m >= 3 - c)

def bfs():
    q = deque([(start, [start])])
    seen = {start}

    while q:
        s, path = q.popleft()
        m, c, b = s

        if s == goal:
            return path

        for x, y in [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]:
            if b == 0:
                nm, nc = m - x, c - y
            else:
                nm, nc = m + x, c + y

            nb = 1 - b
            ns = (nm, nc, nb)

            if (0 <= nm <= 3 and 0 <= nc <= 3
                    and valid(nm, nc) and ns not in seen):
                seen.add(ns)
                q.append((ns, path + [ns]))

    return None

path = bfs()

if path:
    for m, c, b in path:
        print(f"m:{m} c:{c} b:{'L' if b == 0 else 'R'}")
else:
    print("No solution found")
```

Output

```
m:3 c:3 b:L
m:3 c:1 b:R
m:3 c:2 b:L
m:3 c:0 b:R
m:3 c:1 b:L
m:1 c:1 b:R
m:2 c:2 b:L
m:0 c:2 b:R
m:0 c:3 b:L
m:0 c:1 b:R
m:1 c:1 b:L
m:0 c:0 b:R
```